Siddharth Parekh

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EDUCATION

CARNEGIE MELLON UNIVERSITY Pittsburgh, PA

Master of Science in Computer Science (Aug 2026)
Thesis: Mechanisms of Tabular Reasoning in LLMs; Advisor: Prof. Carolyn Rosé

Bachelor of Science in Computer Science, Minor in Machine Learning (May 2025)
University Honors, **School of Computer Science College Honors**
GPA: 3.8/4.0; Dean's List with High Honors: Fall 2021, Fall 2022, Spring 2023, Fall 2024, Spring 2025

Selected Coursework: LLM Systems, Advanced NLP, Computational Game Solving, Deep Learning, Convex Optimization, Probabilistic Graphical Models, Database Systems, Data Structures and Algorithms, Markov Processes

SKILLS

Programming: Python, C/C++, Javascript/Typescript, SQL
ML: PyTorch, Transformers, NumPy, Pandas, Scikit-Learn, Model Training/Fine-Tuning, Evaluation Pipelines
Tools: Git, Docker, AWS, Google Cloud Platform, React, Tailwind, LaTeX

PROJECTS

Discero: Self-Learning Platform (discero.vercel.app, github.com/sparekh9/discero-gen) Jun 2025 – Present

- Built an **agentic course-generation** CLI (**LangGraph**) with a provider-agnostic LLM adapter.
- Developed a paired **React/TypeScript** web app (**Vite, TailwindCSS, Firebase Auth/Firestore**), enforcing a Pydantic/Zod shared JSON schema contract between the generator and frontend for validated course import/export.

Solving Coup (github.com/sparekh9/coup) Oct 2025 – Dec 2025

- Implemented **Counterfactual Regret Minimization** (CFR) solver, applying Discounted CFR with **Monte Carlo sampling** and depth-limited solving to achieve convergence to **near-optimal** strategies for variants of **Coup**, an imperfect information game.
- Achieved **10-100x** performance improvement through systems-level optimizations, including information set encoding via **bit-packing**, templated **C++** architecture with compile-time optimizations, and **strategic pruning** to reduce state space complexity.

EXPERIENCE

Language Technologies Institute, CMU Pittsburgh, PA

Graduate Research Assistant Aug 2025 – Present

- Reverse-engineering the **internal geometry** of structured data processing using linear probes, and activation patching to uncover **coordinate systems** and **variable-binding mechanisms** LLMs use to navigate flat serialization formats (e.g., CSV, Markdown).
- Isolating the **computational circuits** (specific attention heads, MLP layers, etc.) governing **retrieval and reasoning** on tabular data and examining the effects of formatting changes to the serialized string on downstream performance.

Undergraduate Research Assistant Jan 2024 – May 2025

- **SCS Undergraduate Honors Thesis**: Investigated efficient **post-training methods** to emphasize correctness in LLMs for question answering through **Equilibrium Ranking** (reconciling generative and discriminative querying) and leveraging specialized knowledge domains by having the model adopt various **expert personas.**
- Developed evaluation metrics and graph-based models for visually rich document understanding with Prof. Carolyn Rosé and Armineh Nourbakhsh (see **Publications**).

Research Intern Jun 2023 – Aug 2023

- Led research on **numeracy** in LLMs under Prof. Carolyn Rosé, focusing on **financial document parsing** and **question answering** tasks using datasets like FinQA, TAT-QA, and PACIFIC.
- Improved baseline models' accuracy by **3-5%** by integrating **code-generation** LLMs (T5, CodeT5) for numerical reasoning.

PUBLICATIONS

- [**Outstanding Paper**] Z. Wu, R. Dutt, L. M. Breitfeller, A. Nourbakhsh, **S. Parekh**, C. Rosé. R^2-CoD: Understanding Text-Graph Complementarity in Relational Reasoning via Knowledge Co-Distillation. IJCNLP-AACL Main 2025.
- A. Nourbakhsh, **S. Parekh**, P. Shetty, Z. Jin, S. Shah, C. Rosé. Where is this coming from? Making groundedness count in the evaluation of Document VQA models. NAACL Findings 2025.
- A. Nourbakhsh, Z. Jin, **S. Parekh**, et al. AliGATr: Graph-based layout generation for form understanding. EMNLP Findings 2024.